25900283 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCHE200 - INORGANIC CHEMISTRY - I

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. Which of the following is a correct match? [A] / [CO1]

   a). sp^3 – linear b). sp^2 – tetrahedral

   c). sp – linear d). sp^3d^2 – trigonal planar

2. The shared pairs of electrons are called…… [K] / [CO1]

   a). Bond Pairs b). Lone pairs

   c). Non-bonding pairs d). None of these

3. Based on its MO diagram, what is the bond order and magnetic behavior of the N_2 molecule? [A] / [CO1]

   a). Bond order 3, diamagnetic b). Bond order 2, paramagnetic

   c). Bond order 2, diamagnetic d). Bond order 3, paramagnetic

4. Fajan's rule is applicable mainly to [K] / [CO1]

a). Covalent compounds only

b). Ionic compounds with polarizable ions

c). Metals only

d). Noble gases

5. Pitchblende is ---------- [K] / [CO2]

a). UO_2

b). UF_6

c). $ThSiO_4$

d). UCl_4

6. Which actinide ion in +3 oxidation state has just half-filled 5f-subshell? [An] / [CO2]

a). Pu^{3+}

b). Am^{3+}

c). Cm^{3+}

d). Bk^{3+}

7. The carbon -14 dating method is based on the fact that [U] / [CO3]

a). ratio of carbon–14 and carbon–12 is constant

b). carbon–14 is the same in all objects

c). carbon– 14 is highly insoluble

d). All options are correct

8. The oxidation state in Co in $[Co(NH_3)_6]Cl_3$ is [A] / [CO4]

a). +2 b). +3 c). 0 d). +4

9. A chelating ligand must have: [K] / [CO4]

a). One donor atom b). Two or more donor atoms c). No donor atom d). Variable donor atoms

10. [K] / [CO4]

According to Werner's theory, the primary valency of a metal in a coordination compound is always satisfied by:

a). Neutral molecules

b). Negative ions

c). Positive ions

d). Both neutral molecules and negative ions

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11. What are the limitations of Valence Bond Theory? [K] / [CO1]

12. Define hybridization and state two of its key characteristics. [U] / [CO1]

13. Why is lattice energy not measured directly, and how does the Born-Haber cycle help determine it? [U] / [CO1]

14. A molecule of KBr has a bond length of 2.82 A° and an observed dipole moment of 10.41D. Calculate the percentage ionic character of the bond [A] / [CO1]

15. Discuss the effect of lanthanide contraction on separation of lanthanides. [U] / [CO2]

16. Calculate CFSE for a Tetrahedral complex having d^7 electronic configuration [U] / [CO4]

[3x4 = 12]

Part C

Short Essay Questions

Answer any **2** questions

Each question carries **7** marks

17. Derive the Born-Landé equation used for calculating the lattice energy of ionic compounds [A] / [CO1]

18. Discuss the role of chain reactions in nuclear energy. How are they controlled in nuclear reactors? [U] / [CO3]

19. Compare the stability of $[Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$. Explain your answer. [A] / [CO4]

[7x2 = 14]

Part D

Essay Type Questions

Answer any **1** questions

Each question carries **14** marks

20. Define half-life and decay constant. Derive the mathematical relationship between them and discuss their significance in the study of radioactivity [U] / [CO3]

21. Explain the general characteristics of d block elements. [K] / [CO4]

[14x1 = 14]